

Adaptive Unicast-Multicast Strategies for Over-the-Air Updates in Multi-Receiver Wi-Fi Networks

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Abstract—Wireless Over-the-Air (OTA) data delivery in industrial networks must operate reliably across multiple receivers under uncertain channel conditions. This paper studies the problem of efficiently delivering common data to multiple receivers within a limited time horizon by dynamically switching between unicast and multicast transmissions. Based on two complementary optimization objectives reflecting different notions of resource efficiency, we propose three adaptive communication strategies for multi-receiver scenarios. Through extensive simulations under varying channel conditions, we show that adaptive mode selection consistently outperforms static unicast or multicast strategies, with performance gains increasing as the number of receivers grows. For a network with ten receivers, the proposed strategies reduce the total number of transmissions by up to 75% under high multicast reliability and by approximately 25% when multicast reliability is low.

Index Terms—Unicast, Multicast, MDP, Multi-Receiver

I. INTRODUCTION

Industrial Internet of Things (IIoT) networks demand increasingly reliable and efficient approaches for handling large communication volumes [1]. Novel manufacturing approaches, particularly in the context of the Industry 4.0 transformation, rely on efficient and reliable wireless networks to coordinate and optimize work on the assembly line [2]. In this paper, we focus on the specific problem of programming a large number of electronic control units (ECUs) in an automotive factory.

In our focus scenario, multiple ECUs are installed in a vehicle during the assembly process. The system model is shown in Fig. 1. Assembly lines in automotive factories follow strictly regulated processes in which timing constraints are critical. With the ongoing transition toward increasingly smart and flexible assembly processes, wireless technologies are preferred to efficiently transfer the required data to the ECUs, avoiding to support and maintain dedicated wired networks.

Industrial wireless networks [3], however, are typically tailored for a relatively small amount of data and cannot provide the bandwidth required for programming all ECUs in a vehicle within the required time. For example, internal forecasts from BMW indicate that by 2030, more than 30 GB of data will have to be transferred to the different ECUs and, in a single assembly line, tens to hundreds of vehicles can be assembled. Although the overall assembly may take multiple hours, the time allowed for programming all ECUs is much smaller and ranges from about 20 min to 40 min, depending on the specific layout of the factory. This poses the challenge of enabling efficient and reliable communication

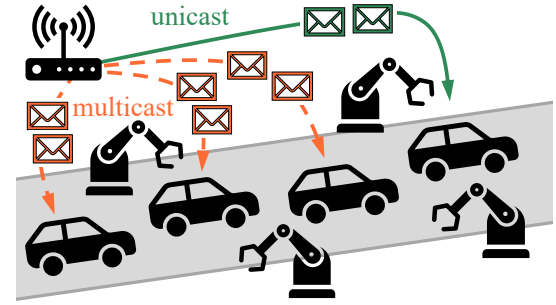


Fig. 1. Programming of ECUs in vehicles that are moving along an assembly line using unicast and multicast transmissions.

of large amounts of data over an industrial wireless network, while the available bandwidth does not meet the necessary performance requirements.

A. Related Work

In [4], Nishida *et al.* study WLANs with parallel unicast and multicast flows and propose a cooperative scheduling method that prioritizes high-rate opportunities and coordinates unicast/multicast transmissions to improve throughput and reduce delay. In [5], Li and Huang study multicast systems with multiple multicast groups over non-identically distributed fading channels and propose a normalized SNR-based fair scheduling scheme that improves fairness among multicast groups while incurring only a small capacity loss compared to greedy scheduling. In [6], Tan *et al.* discuss an opportunistic multicast scheduling in multiple-input multiple-output orthogonal frequency-division multiplexing (MIMO-OFDM) systems, where the access point decides whether to send a unicast or multicast frame. In [7], Mansour *et al.* propose a Groupcast with Retries (GCR) mechanism, where selected receivers provide feedback through a Block Acknowledgment (GCR-BA). It enables the sender to detect missing frames and retransmit the messages when needed. In [8], Yousri *et al.* consider aggregation groups of several frames into a single transmission frame. The authors focus on two main formats defined: MAC Service Data Unit Aggregation (A-MSDU), where all packets are treated as one large frame, and MAC Protocol Data Unit Aggregation (A-MPDU), where each subframe has its own checksum. While aggregation improves efficiency at the MAC layer, it does not address packet-level reliability or adaptive unicast-multicast decision-making for completing data delivery to multiple receivers, which is the

focus of this paper. In [9], Al-Habob *et al.* discuss a deep reinforcement learning (DRL) framework for selecting the best communication strategy between no transmission, unicast, multicast, and broadcast. The goal of this paper is to identify a strategy that minimizes the age-of-information (AoI) and reduces transmit power consumption in a vehicular network. In the context of LTE Multicast-Broadcast Single-Frequency Networks (MBSFNs), Shen *et al.* [10] review the benefit of the MIMO technology for use in unicast and multicast transmissions under a variety of scenarios. They also show that resource efficiency is improved by increasing the number of antennas not only for unicast, but also for multicast.

In summary, the above works mainly focus on MAC-layer scheduling, fairness, and throughput optimization, without explicitly addressing reliable application-layer data delivery to receivers with heterogeneous reception states. In many approaches, multicast feedback is either unavailable or abstracted at the group level, and multicast scheduling approaches typically do not track per-receiver packet reception states or delivery completion. Moreover, while some studies consider switching between unicast and multicast transmissions, they lack a systematic framework that jointly accounts for packet loss probabilities and overall data delivery completion time.

B. Main Contributions

In this paper, we discuss three novel communication strategies implemented at the application layer of the ISO/OSI model, which address limited bandwidth by dynamically switching between unicast and multicast transmission modes. The proposed strategies rely on two specific properties of the considered problem: (i) ECUs share a large amount of common data, such that multiple ECUs can, in principle, receive the same data within a single transmission; (ii) the exact timing and order of programming individual ECUs within the programming interval are irrelevant. The only constraint is that all data must be available to every ECU of a given vehicle before the vehicle reaches a certain point of the assembly line.

Building on these insights, we develop dynamic communication strategies that exploit different notions of efficient resource utilization. Specifically, the strategies are derived from two complementary optimization objectives: maximizing the expected number of successful receptions and minimizing the expected time required to complete all transmissions. The main contributions of the paper are as follows:

- We formulate two complementary optimization problems that capture distinct efficiency objectives for data delivery under a finite time horizon.
- We develop three communication strategies that address these problems under reasonable modeling assumptions and using different solution approaches.
- We evaluate the proposed strategies through extensive simulations under varying system parameters and compare them against pure unicast and multicast baselines.

The following section introduces the system model and optimization problems. Section III presents three solution approaches for these problems. Section IV compares the different

strategies through simulation, and evaluates their performance against a benchmark. Section V discusses a more accurate model of the programming process in a factory and evaluates the three strategies under this refinement. Finally, Section VI summarizes the paper and discusses future research, addressing some of the simplifications considered in this initial work.

II. PROBLEM FORMULATION

This section first introduces the system model and its mathematical representation. It then formulates two optimization problems reflecting different notions of resource efficiency.

A. System Model

In an assembly line, a vehicle under construction moves along a fixed path and is served by different access points (APs) over time. Although multiple ECUs are mounted in a vehicle, typically only one ECU, the *telematic* ECU, is responsible for all communication between the in-vehicle network and external wireless networks. We thus model every vehicle as it had only one ECU, the telematic ECU, and assume that all data required by all other ECUs is received by the telematic ECU.

At any point in time, every telematic control unit (TCU) is connected to a single AP, and we assume that the roaming effects between APs are negligible. We focus on the case of a single AP that concurrently serves a set of vehicles $\mathcal{V} = \{1, 2, \dots, |\mathcal{V}|\}$. All vehicles are assumed to be in coverage of this AP. The system model is illustrated in Fig. 1. We consider a discrete-time approach, where the AP can only send one message per time step. Moreover, at every time step, the AP can select either the *unicast* (targeting a single vehicle) or the *multicast* (targeting multiple vehicles) transmission mode. Each vehicle must receive a subset of the set of messages $\mathcal{M} = \{1, 2, \dots, |\mathcal{M}|\}$ within a finite time horizon N .

Wi-Fi networks rely on a variety of mechanisms to increase the success of communication. We focus on the case where the AP and vehicles rely on the User Datagram Protocol (UDP) for communication. In that case, unicast communication benefits from a retransmission mechanism at layer 2 of the ISO/OSI model, while multicast does not [11], [12]. This implies that unicast communication, also for UDP, can be expected to be more reliable than multicast communication. We consider the different expected reliabilities of unicast and multicast by considering the probabilities p_u (unicast) and p_m (multicast).

To gain analytical insights, we restrict our attention to the case where success probabilities are constant and independent across receivers. Acknowledgment (ACK) and negative acknowledgment (NACK) packets are assumed to be instantaneous and error-free for both unicast and multicast transmissions. Thus, the AP has perfect knowledge about which messages have been successfully delivered to each vehicle, enabling an accurate update of the system state after every transmission attempt.

Let $\sigma(k)$ denote the communication strategy used at time k by the AP. If $\sigma(k) = 1$, then a unicast approach is used. If $\sigma(k) = 0$, then a multicast approach is used. The system state at time k is represented by a matrix $\mathcal{S}(k)_{|\mathcal{M}| \times |\mathcal{V}|} \in \{0, 1\}$,

where $s_{i,j}(k) = 1$ if vehicle j still requires message i , and $s_{i,j}(k) = 0$ otherwise. The set of possible states is given by

$$\mathcal{Z} = \{\mathbf{S}_{|\mathcal{M}| \times |\mathcal{V}|} \in \{0, 1\}\}. \quad (1)$$

The matrix $\bar{\mathbf{S}}_0$, whose elements are all zeros, represents the final target state in which every vehicle has received all the required messages.

At each time step k , the AP selects an action matrix $\mathbf{A}(k)_{|\mathcal{M}| \times |\mathcal{V}|} \in \{0, 1\}$ indicating which message is transmitted to which vehicle(s). The value $a_{i,j}(k) = 1$ means that message i is intended for vehicle j , whereas $a_{i,j}(k) = 0$ indicates that message i is not transmitted to vehicle j at time step k . If $\sigma(k) = 1$ (unicast), then only one element of $\mathbf{A}(k)$ can be nonzero. If $\sigma(k) = 0$ (multicast), then only one row of $\mathbf{A}(k)$ can contain nonzero elements, corresponding to the selected message for that slot. Due to the noisy nature of the wireless communication channel, not all vehicles may correctly receive the intended message. The instantaneous reception outcome is captured by the matrix $\Xi(k)_{|\mathcal{M}| \times |\mathcal{V}|} \in \{0, 1\}$, where each entry $\xi_{i,j}(k)$ is a random variable indicating whether a transmission to vehicle j is successful.

At the beginning of every time step, the AP selects $\sigma(k)$ and an appropriate value for $\mathbf{A}(k)$ based on the constraints set by $\sigma(k)$. The state evolution can then be written as

$$\mathbf{S}(k+1) = \max(\mathbf{0}, \mathbf{S}(k) - \Xi(k) \circ \mathbf{A}(k)), \quad (2)$$

where $\circ$ denotes the element-wise (Hadamard) product.

The size of the state space grows exponentially with the number of messages and vehicles, and we have,

$$|\mathcal{Z}| = 2^{|\mathcal{M}| |\mathcal{V}|}, \quad (3)$$

which poses a challenge for any communication strategy based on state exploration, under typical realistic system dimensions. Nevertheless, the specific problem we discuss allows for a massive reduction of the actual state dimension that a strategy needs to consider. In the simplified setting of our focus, the probabilities of successful message reception under unicast and multicast are constant, and there is no predefined ordering among messages. Furthermore, messages are statistically equivalent from the sender's perspective. This allows the analysis to focus on a single message and consider the state evolution based on the vehicles that still require the single message. This implies that any strategy we consider can focus on a single message at a time, thereby having to cope with a state dimension of $|\mathcal{Z}| = 2^{|\mathcal{V}|}$, which is significantly smaller than (3). Since only one message is taken into account now, the matrix $\mathbf{A}(k)$ is considered along one of its rows only.

We can also observe that once the AP has selected a multicast approach, only addressing all the vehicles that still require the given message is a relevant strategy. Furthermore, in the case of the unicast approach, it is not important which vehicle, among those that still require the relevant message, is selected. For this reason, the selection of the matrix $\mathbf{A}(k)$ reduces to the selection of a value for $\sigma(k)$. If $\sigma(k) = 0$, then all vehicles that still need to receive the relevant message shall be selected. If $\sigma(k) = 1$, then one vehicle, among the vehicles that still need to receive the relevant data is selected, but it is

not important which one is selected.

By relying on the specific simplifying assumptions of the problem, we could reduce the state space of the system and the set of possible actions that the AP can use. In the following, we thus use the integer variable $s(k)$ to denote the system state in time step k , and $\bar{s}_0$ to denote the final target state $s = 0$. For example, if $|\mathcal{V}| = 10$, then $s(k) \in \mathcal{S} = \{0, 1, \dots, 10\}$. Let $\Delta(k) \in \{0, 1, \dots, s(k)\}$ be a random variable denoting the number of successful receivers after an action $\sigma(k)$ is taken. If $\sigma(k) = 1$, $\Delta(k) = \xi_j(k)$ is a random variable depending on the unicast probability p_u . If $\sigma(k) = 0$, $\Delta(k) = \sum_{j=1}^{s(k)} \xi_j(k)$, where $\xi_j(k)$ are independent and identically distributed (i.i.d.) random variables depending on the multicast probability p_m . Then, analogously to (2), the state evolution is written as

$$s(k+1) = \max(0, s(k) - \Delta(k)). \quad (4)$$

Ultimately, this reduction leads to the problem of finding a communication strategy that delivers all the required messages within a finite time horizon by dynamically switching between unicast and multicast transmissions.

B. Optimization Formulation

As the number of transmission opportunities is limited to a finite horizon N , the AP must exploit each time slot as efficiently as possible. To formalize this goal, we formulate two optimization problems that capture different notions of time efficiency. Our first objective is to maximize the total number of successful message deliveries over the time horizon N . To this end, we define a reward function capturing the increase in the number of successfully delivered messages between two consecutive time steps to quantify the instantaneous progress of the system. After applying the state space reduction, the reward is mathematically expressed as

$$r(s(k), s(k+1)) = s(k) - s(k+1), \quad (5)$$

which shows the total number of vehicles still requiring the message. Summing all rewards over the entire time horizon N then accounts for the total number of successful receptions. The corresponding optimization problem is defined as

$$\mathcal{P}_1: \max_{\{\sigma(k)\}_{k=0}^{N-1}} \mathbb{E} \left[\sum_{k=0}^{N-1} r(s(k), s(k+1)) \right], \quad (6a)$$

$$\text{s.t.} \quad \sigma(k) \in \{0, 1\}, \forall k \in \{0, \dots, N-1\}. \quad (6b)$$

The objective is to maximize the expected cumulative reward, with the reward function $r(\cdot)$ defined in (5), and the state evolution $s(k+1)$ given by (4), which depends on the selected action $\sigma(k)$ via the random variable $\Delta(k)$.

A complementary perspective on efficient use of the available horizon is to focus on the time needed to complete the entire update process. Instead of fixing a horizon and maximizing the number of delivered messages, we seek a policy that reaches the all-zero state as fast as possible. Mathematically, the second optimization problem is formulated as

$$\mathcal{P}_2: \min_{\{\sigma(k)\}_{k=0}^{h-1}} \mathbb{E}[h] \quad (7a)$$

$$\text{s.t.} \quad s(h) = \bar{s}_0, \quad (7b)$$

$$h \leq N, \quad (7c)$$

$$\sigma(k) = \{0, 1\}, \forall k \in \{0, \dots, h-1\}, \quad (7d)$$

$$\sigma(k) = \emptyset, \forall k \in \{h, \dots, N-1\}. \quad (7e)$$

where the objective is to minimize the expected completion time $\mathbb{E}[h]$ such that the final target state, where all messages are delivered, is reached. Note that the horizon N must be sufficiently large to ensure feasibility. Transmission strategies are only chosen up to the completion time h (7d), while no transmissions are done for $h \leq k \leq N-1$ (7e).

III. PROPOSED APPROACHES

To identify suitable communication strategies, this section presents three solution approaches for problems $\mathcal{P}_1$ and $\mathcal{P}_2$.

A. A Markov Decision Process Approach for $\mathcal{P}_1$

The system behavior is modeled as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, T, r)$. The *set space* $\mathcal{S}$ is the set of all possible numbers of vehicles that still require the given message. For example, if $|\mathcal{V}| = 10$, then $\mathcal{S} = \{0, 1, \dots, 10\}$. The *action space* describes the possibility to send a message as unicast or multicast. We define $\mathcal{A} = \{\sigma_0, \sigma_1\}$, where σ_0 indicates a multicast approach and σ_1 a unicast approach. With $\sigma(k)$ we denote the action taken in the k^{th} time interval.

The set T represents the probability of transitioning from one element of the state to the next, given a certain action. Let $t(s, s', a)$ be the probability of moving from state $s \in \mathcal{S}$ to state $s' \in \mathcal{S}$ when action $\sigma \in \mathcal{A}$ is taken. If $\sigma = \sigma_0$ (multicast), we have

$$t(s, s', \sigma_0) = \begin{cases} p_m^{(s-s')} \cdot (1-p_m)^{s'} & \text{if } s' \leq s, \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

If $\sigma = \sigma_1$ (unicast), we have

$$t(s, s', \sigma_1) = \begin{cases} p_u & \text{if } s - s' = 1, \\ 1 - p_u & \text{if } s = s', \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

The *reward function* r is defined as the number of messages correctly received from one state to the next, i.e.,

$$r(s, s') = s - s'. \quad (10)$$

The objective of the MDP is to find a stationary policy π that maps each state in $\mathcal{S}$ to an action in $\mathcal{A}$ and maximizes the expected discounted cumulative reward over a finite horizon, expressed as

$$\mathbb{E} \left[\sum_{k=0}^{N-1} \gamma^k r(s_k, s_{k+1}) \right], \quad (11)$$

where γ is the discount factor satisfying $0 \leq \gamma \leq 1$.

B. A Heuristic to Maximize the Number of Received Messages

While the MDP formulation allows identifying an optimal policy for the specific expected reward over a finite horizon, it requires tuning the discount factor in order to identify policies that work as expected in industrial scenarios. Furthermore, its applicability is limited to offline computation, and it does

not easily scale to more complex scenarios in which the probability of successful communication varies over time.

Therefore, the key idea of the second proposed strategy is to estimate, at each time step, the expected improvement resulting from either a unicast or a multicast transmission, and to select the mode that provides the higher expected gain in the number of received messages.

For a unicast transmission, the expected number of delivered messages in a single step is p_u , since only one vehicle is targeted and the success probability is p_u . For a multicast transmission, the same message is transmitted to several vehicles simultaneously, each with an independent success probability p_m . If ω vehicles still require the message, then the expected number of successful receptions is ωp_m .

At each iteration, the AP selects the transmission mode that maximizes the expected successful receptions. We can write

$$\sigma(k) = \begin{cases} 1 & \text{if } \omega p_m \leq p_u, \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

C. A Graph-Based Optimization Solution for $\mathcal{P}_2$

While the previous strategies focused on maximizing the number of successful message receptions, a graph-based optimization approach addressing $\mathcal{P}_2$, i.e., trying to minimize the time (transmissions) needed to successfully transmit all required messages to the vehicles, is developed in this subsection.

Each system state is represented as a vertex, and every feasible transition between two states is modeled as a directed edge (s, s') , where s' denotes the resulting state after a transmission. If a node s' is reachable from s using both a unicast and a multicast transmission, then two edges will connect s to s' . We model the search for the best communication strategy starting from a node $s(0)$ as the search for the shortest path from $s(0)$ to the node representing the state $\bar{s}_0$.

1) Weight for the unicast transmission case

For unicast transmission, we model the edge weight as the expected number of attempts required for successful message delivery, assuming unicast is used until the new state is reached. The expected number of attempts under this assumption is given by $1/p_u$, which follows directly from the geometric distribution.

2) Weight for the multicast transmission case

For a multicast transmission, we model the edge weight as the expected number of attempts required for a successful transition to the new state, assuming only a directed transition between the two states.

Suppose that ω vehicles are selected for multicast communication, and let Δ denote the number of vehicles that successfully receive the message during the transition from s to s' . The conditional probability of transitioning to state s' , given that the system either stays in s or moves to s' is

$$\Pr\{(s, s') | (s, s) \cup (s, s')\} = \frac{p_m^\Delta (1-p_m)^{\omega-\Delta}}{(1-p_m)^\omega + p_m^\Delta (1-p_m)^{\omega-\Delta}} = \frac{p_m^\Delta}{(1-p_m)^\Delta + p_m^\Delta} = p_j. \quad (13)$$

In this expression, the numerator represents the probability of successfully transitioning from state s to s' (i.e., when Δ receivers successfully receive the message), while the denominator normalizes over the two possible outcomes: staying in the same state or transitioning to s' .

Similar to the unicast case, the number of attempts required until the successful transition follows a geometric distribution with success probability p_j . The expected number of attempts is therefore given by $1/p_j$, which after substituting (13) into the latter, reduces to

$$1 + \left(\frac{1 - p_m}{p_m} \right)^\Delta.$$

3) Shortest-Path Formulation

Using these expected times, the state transition graph can be weighted accordingly. With this formulation, the expected completion time for the entire transmission process is given by the total cost of the shortest path from the initial state $s(0)$ to the terminal state $\bar{s}_0$.

4) Analysis of the Estimation

The estimations and weight assignments highlight a trade-off. In the case of pure unicast actions, if there are ω receivers still missing the message, each requires on average $1/p_u$ attempts. Therefore, the expected value of transmissions for reaching all ω receivers is given by ω/p_u . In the case of a multicast action, the expected number of transmissions required to move forward is $1 + (\frac{1-p_m}{p_m})^\omega$.

Fig. 2 compares the expected costs of pure unicast and pure multicast for all values of p_u and p_m , and for different numbers of receivers. The curves in the figure indicate the parameter pair (p_u, p_m) at which the expected number of transmissions for unicast and multicast are equal. Consequently, for a given ω , multicast is more efficient if the point (p_u, p_m) lies above the corresponding curve, whereas unicast is more efficient if it lies below.

For $\omega = 1$, the two communication methods are equivalent since multicast reduces to a single unicast. As ω increases, multicast outperforms unicast at higher values of p_m . Specifically, as $\omega \rightarrow \infty$, the boundary between unicast and multicast approaches $p_m = 0.5$. In general, multicast is consistently preferred when $p_m \geq 0.5$ and $\omega > 2$, independent of the unicast success rate. In contrast, if p_m is small, unicast may still dominate despite its linear scaling with ω .

In the dashed region, multicast is always preferred regardless of the number of receivers ω . In the dotted region, unicast is always preferred for all ω . Along the boundary between the two, the choice depends on ω : for smaller numbers of receivers, unicast is more efficient, while for larger numbers of receivers, multicast becomes favorable. Note that in reality only the points lying below the red line are reasonable, as in general $p_u > p_m$.

IV. SIMULATION RESULTS

This section describes the experiments conducted to evaluate the different strategies and compares the results.

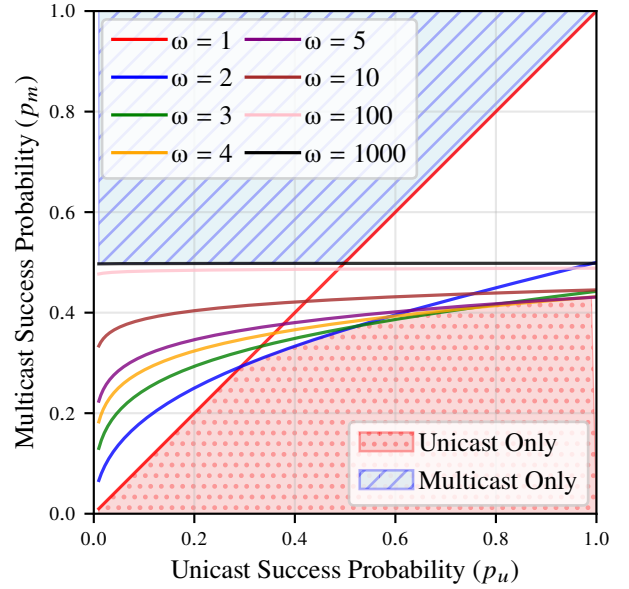


Fig. 2. Selection of unicast and multicast, and equality curves based on the graph-based approach weight assignments.

A. Simulation Setup

The simulation framework was implemented in Python. The simulations were designed to provide a systematic comparison of the proposed strategies. Each simulation run models the process of delivering all required messages to all receivers until the final state is reached or the maximum number of steps is exceeded. The maximum number of transmission steps was set to $N = 1.5 \cdot |\mathcal{V}|$. In practice, fewer than 1% of all simulation runs exceeded this threshold, and the number of undelivered messages was negligible. For the MDP Policy, the discount factor was set to $\gamma = 0.9$.

To provide an insight into the behavior of each strategy under uncertainty and capture the spread and reliability of the results, 10,000 simulation runs were conducted per configuration. The outcomes are represented as histograms. Since the number of transmission steps is discrete, these histograms represent the empirical probability mass function (PMF) of the total time required for completion.

B. Evaluation

Fig. 3 shows the initial set of results. We analyze two cases with different receiver counts:

1) Case 1 - Small receiver set ($|\mathcal{V}| = 3$)

In this case, all strategies converge to similar results, both in terms of completion time and their proportion of unicast and multicast modes. As shown in Figs. 3a) and 3b), when the multicast probability is low ($p_m = 0.2$), only unicast transmissions are selected. When the success probability of multicast increases to $p_m = 0.7$, both unicast and multicast modes are selected.

The empirical PMFs also correspond to theoretical expectations. For example, when $p_m = 0.2$ (Fig. 3a)), all strategies only select unicast, so the probability of completion in three transmissions is $0.9^3 \approx 0.73$, which matches the highest bar on the left. For $p_m = 0.7$, a single multicast can reach all

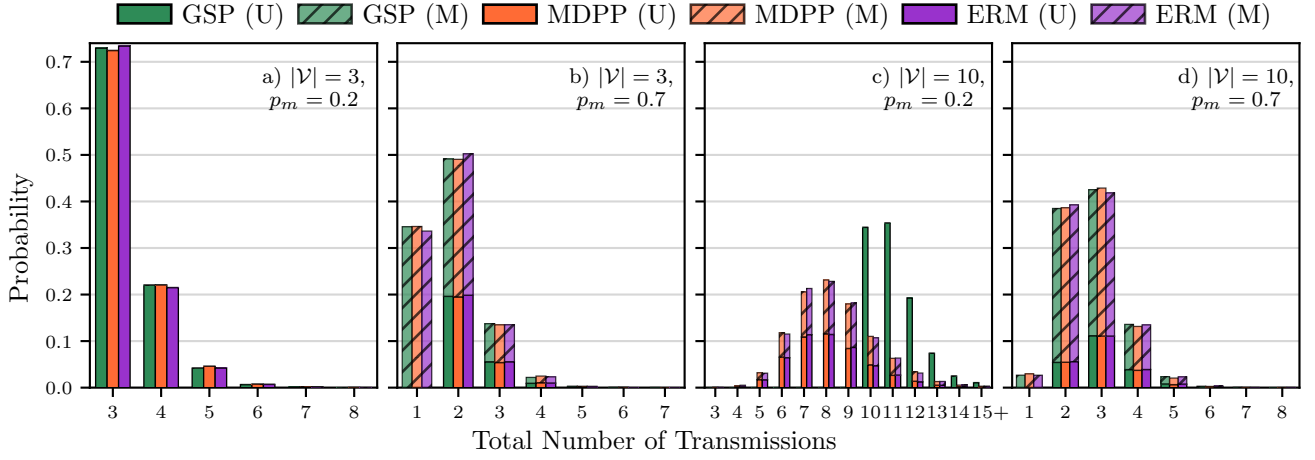


Fig. 3. Empirical Probability Mass Functions (PMFs) of the number of total transmissions (unicast (U) and multicast (M)) for the different strategies: Graph Shortest Path (GSP), MDP Policy (MDPP), and Expected Receptions Maximization (ERM), with $|\mathcal{M}| = 1$, $|\mathcal{V}| \in \{3, 10\}$, $p_u = 0.9$, and $p_m \in \{0.2, 0.7\}$.

receivers. In this case, finishing in a single time step has probability $0.7^3 \approx 0.343$, which is visible in Fig. 3b).

2) Case 2 - Larger receiver set ($|\mathcal{V}| = 10$)

As shown in Figs. 3c) and 3d), when we increase the number of vehicles to $|\mathcal{V}| = 10$, on the one hand, all strategies converge to similar results when the multicast success probability is high ($p_m = 0.7$). On the other hand, for lower multicast reliability ($p_m = 0.2$), the behavior changes, and the empirical PMFs of the strategies differ. In this case, the graph-based approach selects the unicast mode, resulting in a larger completion time because the calculated total weight for the multicast transmissions is larger than that for the unicast transmissions. The MDP policy and the heuristic that maximizes the expected number of receptions outperform the graph-based approach. In these cases, both unicast and multicast modes are selected in the process.

These experiments indicate that in the graph-based approach, there exists a *breakpoint* of p_m at which the preference shifts from multicast to unicast. To determine this threshold more precisely, we examine the expected progress of the two modes when the system is in its initial state, i.e., when $\omega = |\mathcal{V}| = 10$ vehicles still require the message. Let D_u and D_m be the random variables denoting the total number of transmissions required to complete delivery under unicast and multicast modes, respectively. We compare the two expected values of the total number of transmissions, corresponding to

$$\mathbb{E}[D_u] = \frac{\omega}{p_u} \quad \text{and} \quad \mathbb{E}[D_m] = 1 + \left(\frac{1 - p_m}{p_m} \right)^\omega. \quad (14)$$

These values are derived from the edge weights defined in Subsection III-C1 (one edge weight multiplied by ω) and Subsection III-C2 (when $\Delta = \omega$). The multicast mode is advantageous in the initial state if $\mathbb{E}[D_u] > \mathbb{E}[D_m]$, which yields the condition

$$\frac{\omega}{p_u} > 1 + \left(\frac{1 - p_m}{p_m} \right)^\omega. \quad (15)$$

For $p_u = 0.9$ and $\omega = 10$, this inequality can be solved numerically for p_m . The resulting multicast success probability (breakpoint) is approximately $p_m^* \approx 0.43$.

- For $p_m > p_m^* \approx 0.43$, multicast is preferred at the start (similar to the case of $p_m = 0.7$).
- For $p_m < p_m^*$, multicast becomes costly relative to unicast, and the graph shortest path strategy switches to the unicast mode (similar to the case of $p_m = 0.2$).

To validate this analytically derived breakpoint, we conduct additional simulations for $p_m \in \{0.42, 0.43, 0.44\}$ while keeping $p_u = 0.9$. The results in Fig. 4 confirm that the divergence in strategy behavior begins to appear between $p_m = 0.42$ and $p_m = 0.43$, matching the theoretical prediction. For $p_m = 0.42$, the minimum number of transmissions using the graph-shortest path strategy that was observed was 10, indicating that this strategy solely relies on unicast transmissions in this case. For the strategies focusing on maximizing the number of receptions, only 4 or 5 transmissions were required in the majority of the simulation runs. Once p_m increases, the graph-shortest path strategy starts to use multicast transmissions, and the empirical PMFs start to converge, although for $p_m = 0.43$ the strategies that maximize receptions clearly use fewer transmissions on average (4.75, 4.74, and 4.93 for MDPP, ERM, and GSP, respectively). For a multicast probability of $p_m = 0.44$, no clear difference between the strategies can be observed anymore.

C. Analysis for Adaptive vs. Static Strategies

In this subsection, we benchmark the performance of one of the adaptive strategies we developed in Section III, specifically the MDP policy, against static transmission strategies based on pure unicast and pure multicast under different system configurations and channel reliability settings. The corresponding results are represented in Fig. 5.

We first consider the case with a small number of receivers ($|\mathcal{V}| = 3$). When the multicast success probability is low ($p_m = 0.2$), the MDP strategy converges to pure unicast. This is reflected in the mean completion times of the MDP strategy and the pure unicast scheme (3.34 and 3.33 transmissions, respectively). In contrast, the pure multicast scheme performs worse, requiring 8.69 transmissions on average with a larger variance and a maximum number

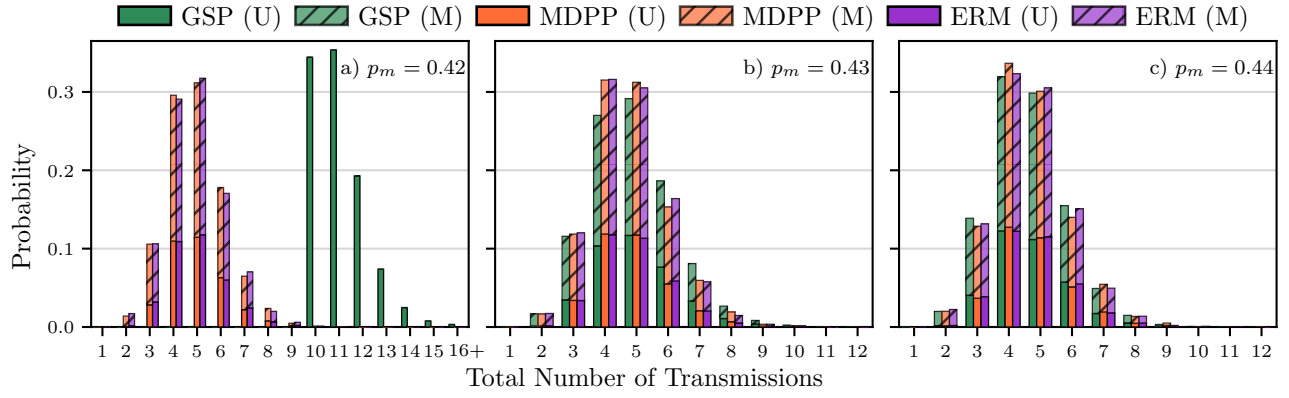


Fig. 4. Empirical Probability Mass Functions (PMFs) of the number of total transmissions (unicast (U) and multicast (M)) for the different strategies: Graph Shortest Path (GSP), MDP Policy (MDPP), and Expected Receptions Maximization (ERM), with $|\mathcal{M}| = 1$, $|\mathcal{V}| = 10$, $p_u = 0.9$, and $p_m \in \{0.42, 0.43, 0.44\}$.

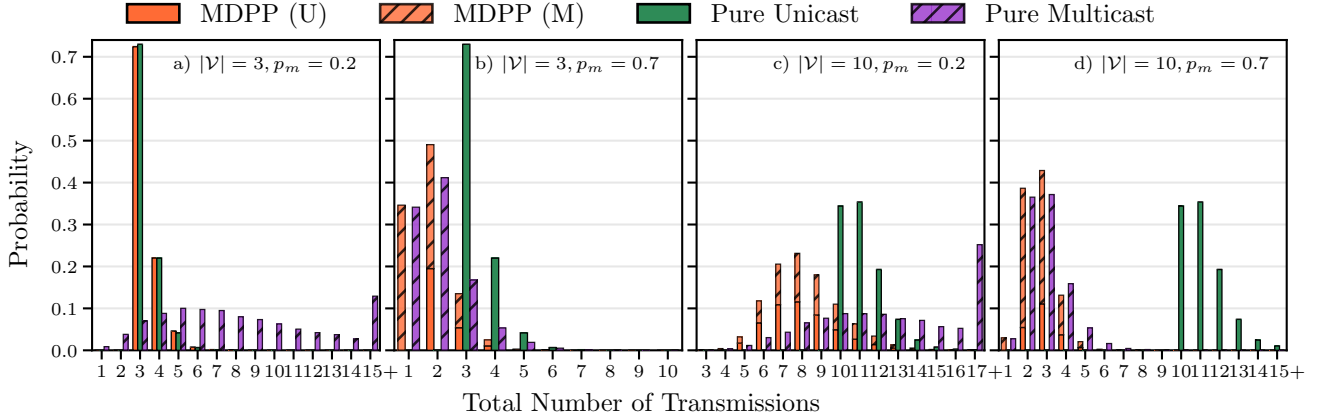


Fig. 5. Empirical Probability Mass Functions (PMFs) of the number of total transmissions (unicast (U) and multicast (M)) for the different strategies: MDP Policy (MDPP), pure unicast, and pure multicast, with $|\mathcal{M}| = 1$, $|\mathcal{V}| \in \{3, 10\}$, $p_u = 0.9$, and $p_m \in \{0.2, 0.7\}$.

of 46 observed transmissions. When the multicast reliability increases to $p_m = 0.7$, the MDP strategy begins to select multicast in the first system state. In this configuration, the adaptive policy reduces the expected number of transmissions to 1.85, whereas pure multicast and pure unicast require 2.02 and 3.33 transmissions, respectively. These results indicate that even with a small receiver set, the adaptive strategy performs better when multicast becomes relatively reliable.

For a larger receiver set ($|\mathcal{V}| = 10$), the gap between the adaptive and static strategies is more evident. When $p_m = 0.2$, pure unicast requires 11.12 transmissions on average, and pure multicast performs worse (13.70) due to frequent transmission failures. The MDP policy achieves a mean of 8.29 transmissions, corresponding to a reduction of about 25% compared to pure unicast. In this setting, the adaptive policy first selects multicast to serve a larger set of receivers. As the number of receivers requiring the message decreases, the policy begins to select unicast to ensure reliability. When multicast reliability increases to $p_m = 0.7$, the adaptive strategy again provides the best performance. The MDP policy requires only 2.74 transmissions on average, compared to 2.93 for multicast and 11.12 for pure unicast. Overall, across all tested configurations, the results consistently show that the adaptive strategy either matches or outperforms the static schemes.

V. REFINED TIMING MODEL

In previous sections, it was assumed that both multicast and unicast transmissions require the same fixed time step. This assumption overlooks the link-layer retransmissions in the unicast approach. In practice, unicast transmissions benefit from lower-layer reliability mechanisms, such as Hybrid Automatic Repeat reQuest (HARQ) [13], which do not exist in the multicast approach. These mechanisms impact both end-to-end delay and reliability. In this section, we refine the system model to account for these mechanisms, providing a more realistic representation of the communication system.

To quantify this effect, we consider a unicast link with a packet error rate q and a maximum of $K = 3$ HARQ retransmissions. The probability of successful delivery is then given by $P_{\text{succ}} = 1 - q^{K+1}$. For a moderately reliable link with $q = 0.1$, this results in $P_{\text{succ}} = 1 - 0.1^4 = 0.9999$, while for a less reliable link with $q = 0.2$, the probability still remains high at $P_{\text{succ}} = 1 - 0.2^4 = 0.9984$. The expected number of transmission attempts can be approximated by

$$t_u^{0.1} = \sum_{n=1}^{K+1} n q^{n-1} (1-q) \approx 1.11 \text{ and } t_u^{0.2} \approx 1.25 \quad (16)$$

for $q = 0.1$ and $q = 0.2$, respectively. These results indicate that the HARQ mechanisms yield high reliability at the cost of an increase in transmission time of around 10–25% compared

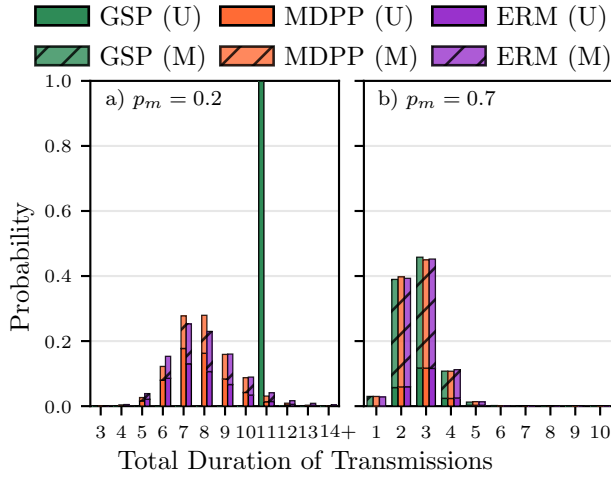


Fig. 6. Probability histogram with the bin size 1 and floats floored to the nearest integer of the total transmission durations for the different strategies: Graph Shortest Path (GSP), MDP Policy (MDPP), and Expected Receptions Maximization (ERM) using the refined timing model for $|\mathcal{V}| = 10$, $p_u = 0.9999$, and $p_m \in \{0.2, 0.7\}$.

to a single multicast attempt.

In the following, the completion times are still from a discrete set but no longer integer-valued. A completion time has the form $x+1.11y$, where x counts multicast transmissions and y counts unicast transmissions. As a result, the outcome space consists of values such as 1.11, 2.11, 3.33, etc. Therefore, instead of plotting PMFs, we use probability histograms to show the empirical distribution of completion times. In the refined model with an error probability of $q = 0.1$, the unicast success probability increases from $p_u = 0.9$ in the simplified model to $p_u = 0.9999$, while the unicast transmission duration increases by 11%. Fig. 6 shows the results for $p_u = 0.9999$ and $|\mathcal{V}| = 10$ under two different multicast reliabilities, using a bin size of 1, and flooring float values to the nearest integers.

In the simplified model shown in Fig. 3c), the graph-based model relies only on unicast, so the total number of transmissions starts with 10 for $|\mathcal{V}| = 10$ receivers. When the unicast transmission time increases by 11%, this start value shifts accordingly to around 11, which is visible in Fig. 6a). As GSP relies solely on unicast transmissions for $p_m = 0.2$ and the success probability $p_u = 0.9999$, the probability for a total duration of around 11 is very high. Nonetheless, observing higher times than 11 is still possible. For the MDP policy and the ERM heuristic, the shift is smaller. In these policies, a fraction of the transmissions are multicast, so the effective increase is less than 11%. A similar behavior can be observed for $p_m = 0.7$. In the simplified model shown in Fig. 3d), all methods require approximately 2.75 transmissions on average. In the refined model, this value even decreases to around 2.7 due to the increased reliability of unicast transmissions. Hence, the majority of vehicles can be served using multicast, while the remaining vehicles are updated using reliable unicast.

VI. CONCLUSION

This work developed and evaluated adaptive strategies for switching between unicast and multicast transmissions to

improve the efficiency of over-the-air updates in multi-receiver wireless networks. Based on the defined optimization problems, three solutions were introduced. These strategies were evaluated under both simplified and refined timing models. The results consistently show that adaptive mode selection outperforms static unicast or multicast, particularly in larger networks and under smaller multicast reliability. Future extensions include introducing time-varying channel models to replace the assumption of fixed success probabilities. Furthermore, introducing a priority structure or a weighted objective would allow the decision process to favor specific receivers. Finally, in the multicast mode, feedback mechanisms are not implemented in practice. Introducing a feedback mechanism would significantly improve the model, allow more precise mode switching, and reduce unnecessary retransmissions.

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